



Text and Audio Framework for an Effective Mathematical Symbols' Visualization for Blind Students

Anshu Singla*

Chitkara University Institute of Engineering and
Technology, Chitkara University, Punjab, India
anshu.singla@chitkara.edu.in

Vandana

Chitkara University Institute of Engineering and
Technology, Chitkara University, Punjab, India
vandana1004cse.phd@chitkara.edu.in

ABSTRACT

There is no denying with the fact that the vision impaired candidates/blind candidates were allocated to explicit instructive settings depending on their impairment. India's schooling framework rebuilt in 2020 brought some major and crucial changes in the curriculum, including the absorption of a language of comprehensive training in approach drives. The instructive encounters of visually impaired/vision-weakened individuals, the elements that impacted these encounters and how these aided or blocked their future odds were the critical focal point of the review and execution of new methods on which this research is based. The blind students or the visually impaired are dependent on two senses, the touch sense and the auditory sense, for their academics and their daily activities. During Pandemic-2019, the importance of auditory sense becomes much more important than the touch sense for the visually impaired people. Online studies for blind students especially subjects like mathematics and science emerged as a big challenge for the society. The researchers need to find effective solution to teach blind students through audio and visual aids. This paper focus on the framework for the effective mathematical symbols visualisation and has been designed to assist students and the teaching staff in text as well as audio format.

CCS CONCEPTS

• : • **Computing methodologies** → Machine learning;

KEYWORDS

Braille, Mathematical Symbols, Braille Mapping, Vision Impairment, Braille Visualization

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*Corresponding author

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1 INTRODUCTION

Nature gifted mankind eyes to improve and appreciate the beauty of the world, so that one and all might get familiarity of nature's knowledge. Education, on the other hand, aids human development in all the aspects in society and in the workplace. Education transforms a person's handicap into potency; vision-impaired people must obtain a high degree of education in order to increase more self-assurance, security, and financial immovability in the society. Blind individuals are educated using Braille system which is a code-based system. Mathematics has always been a complicated subject to teach and learn for blind students. It becomes more challenging when one needs to learn through code-based language. When higher education begins, a huge number of blind students leave away mathematics. The fundamental reason is mathematics becomes more complex and there is very less number of teaching aids available in all over the world, blind students go round to substitute regulation offered by various state boards. Consequently, the blindest students send away mathematics as a workable option. Varied boards have different subject assortment and made mathematics as an optional course. The lack of graphical demonstration of symbols for blind students is the most significant impediment for primary and secondary mathematics teaching [1]. As a result, providing options against mathematical subjects is a standard choice of blind schools for vision-impaired students. Secondly, other optional choices are frequently preferred by teachers and parents over maths. There are extreme requirements for academics, scientists, and the all-purpose public to work on the creation of tools and technology [2, 3] so that vision-impaired people can achieve better level of education, particularly in mathematics. As the aptitude to see one's own future is a crucial constituent of each transition stage [4-7]. The interviewing and selection process become cumbersome if aptitude, graphical representation, and mathematics is not included in professional career [8]. The mathematics curriculum is primarily stressful for blind/vision-impaired people and that accomplishment in particularly mathematics subject is poor for vision impaired candidates [9-11]. Thus number of blind/vision-impaired people progressing to third-level education reduces at large scale. Because mathematics is for the most part visual, two-dimensional, and non-linear in nature, it can be difficult to present [12]. As a result, blind or vision-impaired people may face difficulties. Confusion with Braille notation gets compounded as majority of the mathematics instructors are unfamiliar with mathematics notation [13]. This results in difficulties faced by blind/vision-impaired pupils while interacting with the mathematics syllabus. Classified graphical-spatial mathematics concept, such as tables and graphs, is particularly difficult as Braille has large number of mathematical notations [14]. Furthermore, while speech synthesisers are favourable while using

word-processing packages, they don't always go well when it comes to mathematics [15-18]. The society and education researcher need to pay attention to IoT and cloud based automated online tools that are friendly to blind students [19, 20]. This will improve the quality education provided to visually impaired community. The main objective of this paper is to make blind students well versed with the use of this new technique so that mathematics no longer remain a burden for them rather they should enjoy it like other subjects.

2 LITERATURE REVIEW AND ANALYSIS

During the visits to the blind schools, it was observed that in the early stages of education, blind students were following the touch sense only [21]. They were taught on thermocol and other objects to touch and visualise [22]. For example, in SST subject, to make students understand the map of India, thread is used as shown in Figure 1.



Figure 1: Touch Map to teach blind people

In making them understand the different states, the thread of different thickness is used. This is the most convenient and practical method of teaching the blind students. The students learn and follow the touch techniques [23]. Their IQ level enhances and they understand and follow the instructions in their academics. But when they reach the secondary stages of academics, the technique changes from touch to code. The entire learning criteria changes for a student. This becomes complicated and confusing for the students. When they reach the higher classes, they start opting for different vocational courses like music. They want to pursue their career in the field of science and mathematics, but unfortunately, they don't. They find these two subjects tough and confusing in the higher classes. During the school visit, randomly 10 students were selected and were asked upon their interests in different subjects. Alarming results were observed as shown in Table 1. It can be observed from table 1 that almost no student is interested in learning mathematics. On probing further, the students expressed their helplessness in the mathematics subject. This research main focus is to develop a framework to ease the learning of visual-impaired students who wish to learn mathematics. The main problem identified was the

Table 1: Interest of visually impaired students based on subjects (Total number of students=10)

S.No	Subject	Number of interested Students	Number of not interested Students
1.	Hindi	8	2
2.	Punjabi	7	3
3.	Music	9	1
4.	Social Studies	8	2
5.	Mathematics	2	8
6.	Science	3	7

code perception and visualisation. Therefore, a new code was generated. Effort was made to help students in touch as well audio mode for the blind students, especially in mathematics to make students understand and visualise the symbols easily.

3 A FRAMEWORK FOR VISUALIZATION OF MATHEMATICAL SYMBOLS IN TEXT AND AUDIO

An exact scaffold has been designed for visualization of mathematical symbol for the blind students that facilitates study of mathematics for them. This framework has been focused on different type of mathematical symbols. The framework will work on both texts as well as audio output as shown in Figure 2. Firstly, the user will give the input. It will check whether it is text or audio. If the input is not

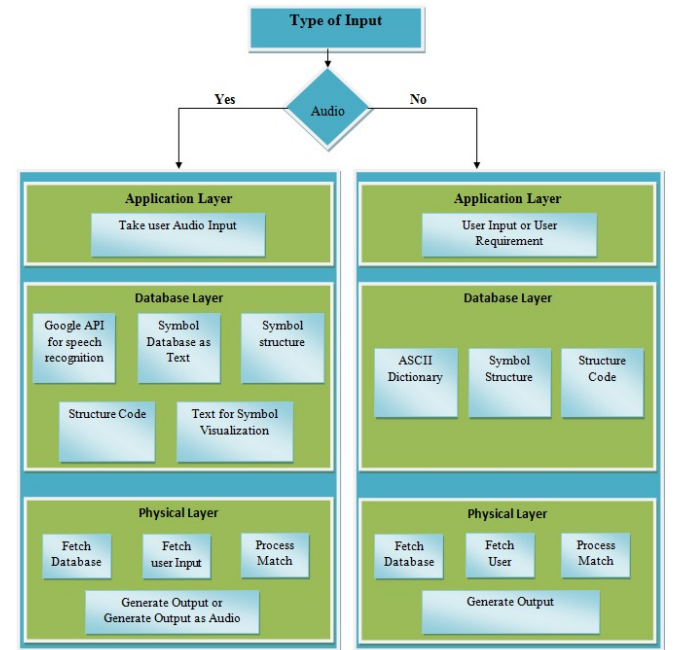


Figure 2: Framework for Visualization of Mathematical Symbols in Text & Audio

audio means it is in text form then right-side framework will work. If the input is in audio then left side framework will work. Both the framework will contain three layers a) Application Layer b) Database Layer c) Physical Layer. The working of both frameworks will describe below.

3.1 TEXT INPUT WORKING

3.1.1 Application Layer. Application layer is based on user input. In this layer, user will give the input that will be in the form of mathematical symbols, text as shown in Figure 3, so that he/she may get output at the later stage in the form of shape of mathematical braille symbol that is called visualization of new braille code.

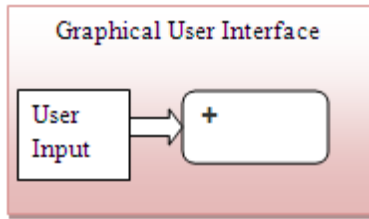


Figure 3: Graphical User Interface for Mathematical Symbol

3.1.2 Database Layer. It includes the existing codes of mathematics symbols presently taught to blind students. It also includes the data of newly mapped visualized mathematical symbol that is the significant part of this research. These symbols have been devised to address the identification of the shape of symbol so that it can be easily visualized in the brain of a blind student. It consists of three parts: a) ASCII Dictionary, b) Symbol Structure, and c) Structure Code explained subsequently.

ASCII Dictionary. First part of this layer is ASCII dictionary. The function of which is to convert input into ASCII code. It will also identify if the ASCII code of given input is present in the ASCII dictionary or not as shown in Figure 4.

S.No	Symbol	ASCII Dictionary
1	<	3C
2	=	3D
3	>	3E
4	α	03B1
5	β	03B2
6	γ	03B3
7	/	2F
8	\	5C
9	[	5B
10	]	5D

Figure 4: ASCII Dictionary for Mathematical Symbol

Symbol Structure. Second part of this layer is symbol structure. The main function of symbol structure is to recognize if the symbol structure of input is present in the database in the form of 2x3 as shown in Table 2. As the braille code follows 2X3 format, therefore effort was made to follow 2x3. The Symbol Structure contains mathematical symbol demonstration and unique code. Mathematical symbol representation describes the touching symbol is mathematical symbol and unique code describes the mathematical symbol category.

Table 2: Symbol structure

S.No	Symbol	Name	Mathematics SymbolRepresentation	Unique Code
1	≡	Equivalence	⠠⠠⠠	⠠⠠⠠
2	<	Less than	⠠⠠⠠	⠠⠠⠠
3	>	Greater than	⠠⠠⠠	⠠⠠⠠
4	≤	Less than equal to	⠠⠠⠠	⠠⠠⠠

Structure Code. Third part of database layer is structure code. The rationale of which is to conform the presence of new visualized Braille code in the form of 2x3 in database as shown in Table 3.

Table 3: Structure Code

S.No	Symbol	Name	Visualization of New Braille Code
1	≡	Equivalence	⠠⠠⠠
2	>	Greater than	⠠⠠⠠

3.1.3 Physical layer. The Next part of framework is physical layer that includes fetch database, fetch user input, match process and generate output. The purpose of fetch database is to bring the data that has already been stored in fetch user input and finally that will be matched by process and generate output as shown in Figure 5. Physical layer storage contains symbol ASCII code, Braille dots, Braille Meaning, Unicode and Braille.

S.No	Symbol	ASCII Dictionary	Braille dots	Braille Meaning	Unicode	Matrix
1	<	3C	2,4,6	gh	2823	[[[01, 0], [1, 0], [0, 1]]]
2	>	3E	1,3,5	ar	281c	[[[1, 0], [0, 1], [1, 0]]]

Figure 5: Physical Layer Storage

Experimental Results. If the input is + then database later firstly will convert that into ASCII code, confirm the symbol structure and finally the presence of structure code that will be stored in to memory called physical layer. The expected output of this framework is shown in Figure 6. The output will be based on touch as well as audio.

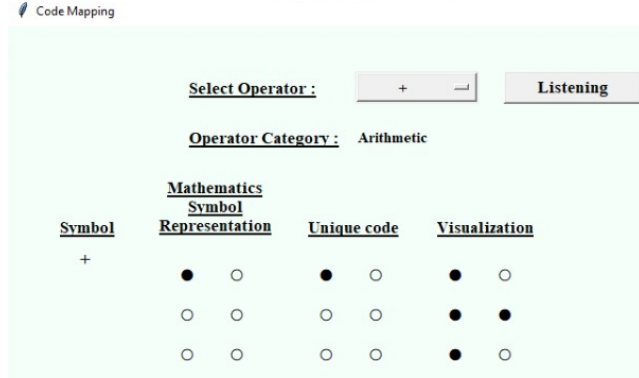


Figure 6: Output of mathematical Symbol

- The first 1 raised dot on the top in the left column will instruct the blind students in audio that the next code they are going to learn is mathematics.
- Next the students will get the audio instruction that the next category is unique code and the code is 1. They will easily perceive that they are following arithmetical category of symbols.
- Finally, the students will get the audio instruction of visualization of new Braille code that is 1-2-3-5. The students will be able to perceive, visualize and judge the symbol easily.

3.2 TEXT INPUT WORKING

3.2.1 Application Layer. Application layer is based on user input in the form of audio. In this layer user will give the input that will be in the form of mathematical symbols, audio as shown in Figure 7.

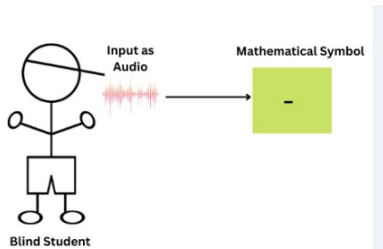


Figure 7: Audio Graphical User Interface for Mathematical Symbol

3.2.2 Database Layer. The five components are present. Additionally, it contains code-based audio data for a newly visible mathematical symbol. These symbols have been coded to address how to identify the symbol's shape so that it can be quickly understood.

Google API for Speech Recognition. This is used to recognize the input because the blind student will give input in the form of speech. It will identify the sound and send to database.

Symbol Database as Text. The main objective of symbol structure as text is to check whether the database has the input's symbol structure as a 2x3 numeric code or not, as shown in Table 4.

Table 4: Symbol Database as Text for Audio

S.No	Symbol	Name	Mathematics Symbol Representation	Unique code	Visualization
1	Γ	Gamma	1	[1-4]	[1, 3, 4, 6]
2	Δ	Delta	1	[1-4]	[2-6]

Symbol Structure. The primary purpose of symbol structure in audio is to determine whether the database has the input's symbol structure as a 2x3 numeric code. As a result of the blind students hearing the code and recognizing the shape of symbol.

Structure Code. Next part of database layer is structure code, which is to confirm the presence of new visualized Braille code in the form of numeric like braille structure of 2x3 in database.

Text for Symbol Visualization. This is how the symbol actually looks. This will produce a numerical output. Student makes the identification of new braille mapped object as shown in Table 5, because braille language is a code language through numeric code.

Table 5: Visualization of Text in the form of Audio

S.No	Symbol	Name	Visualization of New Braille Code
1	ω	Omega	[3, 6]
2	ψ	Psi	[2, 5]

3.2.3 Physical layer. The Next part framework is physical layer that includes fetch database, fetch user input, match process and generate output in the form of audio.

4 CONCLUSION

In this research paper, the work has been done to make the mathematical symbols short, both in the form of text and audio and more comprehensible for the blind students, so that they may get motivated to learn mathematics using braille. This new technique of text and audio mode will facilitate the similar visualization of mathematical symbols in braille as those look originally. The technique will be more beneficial for the students who became visually impaired gradually and completely. The new representation will also

open doors to make their career in the field of mathematics. This new technique is a new step in the direction of attainment of the destination of mathematics for the blind students. The Blind/vision-impaired persons have an extended history of exclusion and missed viewpoint in all areas of schooling.

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